



Charu Karakkaparambil James

📍 **Home** : Altenwoogstraße 27, 67655, Kaiserslautern, Germany

Date of birth: 09/12/1991 **Place of birth**: Ernakulam, India **Nationality**: Indian

EDUCATION AND TRAINING

[01/12/2025 – Current]

PhD in Informatics

RPTU University Kaiserslautern-Landau

City: Kaiserslautern | **Country**: Germany

My PhD research focuses on neural topic modelling for multilingual, low-resource, and temporally evolving text, including cross-lingual knowledge distillation, semantic propagation in medical language, and LLM-based evaluation of dynamic topic models. I also investigate how lexical change can affect traditional topic evaluation metrics and explore semantic alternatives for more reliable assessment.

[01/04/2017 – 31/01/2021]

MSc in Informatics

Technische Universität Kaiserslautern

City: Kaiserslautern | **Country**: Germany

Specialization in **Intelligent Systems**, covering machine learning, AI, deep learning, computer vision, document analysis, and multimedia data mining, with additional studies in software quality and language processing. Master's thesis on **translation-invariant object detection using CNNs**.

[01/09/2009 – 01/04/2013]

BTech in Computer Science and Engineering

Cochin University of Science and Technology

City: Ernakulam | **Country**: India

Bachelor's in **Computer Science and Engineering**, covering programming, algorithms, databases, operating systems, networks, software engineering, AI, security, and distributed/mobile computing, with practical experience through labs and academic projects.

WORK EXPERIENCE

RPTU University Kaiserslautern-Landau

City: Kaiserslautern | **Country**: Germany

[01/03/2021 – 30/11/2025]

Research Assistant

Conducted research in **neural and dynamic topic modelling, continual learning, and LLM-based NLP**, including evaluation of dynamic and hierarchical topic models, development of the **Continual Neural Topic Model (CoNTM)**, cross-lingual knowledge distillation for continual topic modelling, and semantic evaluation methodologies.

KPMG

City: Saarbrücken | **Country**: Germany

[01/08/2019 – 31/01/2021]

Intern and Working Student

Gained insight into general consulting operations within the Lighthouse – Center of Excellence for Data and Analytics. Worked under professional supervision as part of

project teams, primarily focusing on **training deep learning models for extracting invoice line items from invoices**.

SAP

City: Rot-Malsch | **Country:** Germany

[01/05/2018 – 28/02/2019] **Working Student**

Developed a **cloud-based application** within SAP Mission Critical Support, covering requirements analysis, solution design, and implementation using **SAP Web IDE, SAPUI5, HTML, JavaScript, CSS, and REST/OData services**.

Tata Consultancy Services

City: Ernakulam | **Country:** India

[01/07/2013 – 28/02/2017] **Systems and Test Engineering**

management, source control, data management, and test automation.

Software testing and quality assurance across healthcare, insurance, and energy projects, including **requirements analysis, test planning, test case design, functional and integration testing, and test execution**, using **TFS, ALM, SQUIDS, and HP QuickTest Professional**.

SKILLS

C, C#, JavaScript, HTML/CSS, Java | Python (computer programming) | AI platforms(PyTorch, Tensorflow, TensorRT, ONNX Runtime) | Torch, Keras, Tensorflow, Matplotlib, Numpy, Scikit-learn

LANGUAGE SKILLS

Mother tongue(s): Malayalam

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

German

LISTENING A2 **READING** A2 **WRITING** A2

SPOKEN PRODUCTION A2 **SPOKEN INTERACTION** A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PUBLICATIONS

[2026] **Continual Neural Topic Model**

Authors: Charu Karakkaparambil James, Waleed Mustafa, Marcio Monteiro, Marius Kloft, and Sophie Fellenz | **Journal Name:** Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)

[2026] **Semantic Echo Pathways: Tracing How Medical Language Propagates and Transforms**

Authors: Charu Karakkaparambil James, Marcio Monteiro and Sophie Fellenz | **Journal Name:** In Proceedings of the 1st Workshop on Linguistic Analysis for Health (EACL 2026)

[2026] **PROMPT-DTM: A Semantic Evaluation Framework for Dynamic Topic Models using Prompted Large Language Models**

to appear

Authors: Charu Karakkaparambil James | **Journal Name:** IEEE International Conference on Foundation and Large Language Models (FLLM 2026)

- [2026] **MuLTe: Do LLMs Judge Topic Coherence Equally Across Languages? A Human-Grounded Evaluation**
to appear
Authors: Charu Karakkaparambil James | **Journal Name:** IEEE International Conference on Foundation and Large Language Models (FLLM 2026)
- [2026] **When Lexical Change Misleads: Rethinking Dynamic Topic Model Evaluation with Traditional and LLM-Based Metrics**
to appear
Authors: Charu Karakkaparambil James | **Journal Name:** NeurIPS 2026 Workshop WiML
- [2024] [**Evaluating Dynamic Topic Models**](#)
Authors: Charu Karakkaparambil James, Mayank Nagda, Nooshin Haji Ghassemi, Marius Kloft, and Sophie Fellenz | **Journal Name:** Association for Computational Linguistics (ACL 2024)
- [2024] [**Characterizing text datasets with psycholinguistic features**](#)
Authors: Marcio Monteiro, Charu Karakkaparambil James, Marius Kloft, Sophie Fellenz | **Journal Name:** Findings of the Association for Computational Linguistics (EMNLP 2024)
- [2021] [**Hierarchical topic evaluation: Statistical vs. neural models**](#)
Authors: Mayank Nagda, Charu James, Marius Kloft, Sophie Fellenz | **Journal Name:** Proceedings of the Bayesian Deep Learning Workshop, NeurIPS, Virtual
- [2021] [**Provable Translational Robustness For Object Detection With Convolutional Neural Networks**](#)
Authors: Axel Vierling, Charu James, Karsten Berns, Nikoletta Katsaouni | **Journal Name:** 2021 IEEE International Conference on Image Processing (ICIP)